

Convention: power rails above components; GND below.

Components and connections:

- Power Regulation:**
 - U1: 3.3V Regulator (AMS1117-3.3)
 - U2: 5V Regulator (AMS1117-5.0)
 - C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C48, C49, C50, C51, C52, C53, C54, C55, C56, C57, C58, C59, C60, C61, C62, C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C86, C87, C88, C89, C90, C91, C92, C93, C94, C95, C96, C97, C98, C99, C100: Capacitors for decoupling and filtering.
- Microcontroller and Memory:**
 - U3: ESP32-S3-WROOM-1-N16R8
 - U4: PSRAM
- Display:**
 - U5: Display Driver (SN65HVD3082E)
 - U6: 128x64 OLED Display
- Buttons:**
 - BTN1, BTN2, BTN3: Push buttons connected to the microcontroller.
- Connectors and Headers:**
 - J1: RJ45 Ethernet connector.
 - J2: PD, FFC_24 connector.
 - J3: UART-DBG connector.
 - J4: USB-OTG connector.
- Other Components:**
 - R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42, R43, R44, R45, R46, R47, R48, R49, R50, R51, R52, R53, R54, R55, R56, R57, R58, R59, R60, R61, R62, R63, R64, R65, R66, R67, R68, R69, R70, R71, R72, R73, R74, R75, R76, R77, R78, R79, R80, R81, R82, R83, R84, R85, R86, R87, R88, R89, R90, R91, R92, R93, R94, R95, R96, R97, R98, R99, R100: Resistors for pull-up, pull-down, and current limiting.
 - D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21, D22, D23, D24, D25, D26, D27, D28, D29, D30, D31, D32, D33, D34, D35, D36, D37, D38, D39, D40, D41, D42, D43, D44, D45, D46, D47, D48, D49, D50, D51, D52, D53, D54, D55, D56, D57, D58, D59, D60, D61, D62, D63, D64, D65, D66, D67, D68, D69, D70, D71, D72, D73, D74, D75, D76, D77, D78, D79, D80, D81, D82, D83, D84, D85, D86, D87, D88, D89, D90, D91, D92, D93, D94, D95, D96, D97, D98, D99, D100: Diodes for protection and signal conditioning.

Legend:

- #FLG1: Power Flag
- PWR_FLAG: Power Flag
- V12_CAT5E: CAT5E Ethernet cable
- V12_PROT: Protection
- V3V3: 3.3V supply
- V5V5: 5V supply

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Title: Volthium reader – display side

Size: A3 Date: 2026-05-24 Rev: CP-schematic-1d: 1/1
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